6.3 Haar Features for Face Detection Self-Check Quiz

coach

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Assignment details

Due Oct 28, 11:59 PM IST   **Attempts** Unlimited   **Try again**

Submitted Oct 21, 1:57 AM IST

Your grade

To pass you need at least 80%. We keep your highest score.

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1. Which of the following are useful features for face detection? **1 / 1 point**

- ○ Edges
- ○ Corners
- ○ SIFT features
- ◉ None of the above

Correct
As there are too many complex edges and corners in a face image hence they're not a good metric. SIFT features are mostly used to finding matches between two images, not really helpful in face detection.

2. Statement:- Given an image and its reflection along the vertical axis. If a Haar filter is applied to them, we should be getting the same output for both the images. **1 / 1 point**

Determine if the above statement is True/False?

- ○ True
- ○ False
- ◉ Not enough data

Correct
It can be true if the images are symmetric along the vertical axis (i.e. reflection is same as original) and vertical Haar filters are applied. If image. In all other cases it is False.

3. Given there are four Haar filters of size 8 x 8 being applied to an image of dimension 1920 x 1080 pixels. Assume a standard computer processor can perform a million additions/subtractions per second. How long will a computer take to perform the entire operation? **1 / 1 point**

- ○ 2m 04s
- ○ 2m 11s
- ◉ 8m 43s
- ○ 8m 51s

Correct
As seen in lecture, the computation cost of a haar filter is (N x M - 1) additions per pixel per filter per scale. As only one scale is given so we can ignore that. We've N, M = 8, hence total no. of additions is (8 x 8 - 1) x (1920 x 1080) x 4 = 522.55 x106 additions. Hence total time taken is 522.55s or 8m43s.